Sentiments in Hotel Reviews

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***Abstract*—This paper presents a sentiment analysis of hotel reviews using a BERT-based model to classify reviews as positive or negative. The dataset, sourced from Hugging Face, includes hotel reviews and the model aims to provide actionable insights on customer sentiment. By analyzing the distribution of positive and negative reviews over time, we identify trends and shifts in consumer opinions. Despite the high accuracy of BERT, the model's computational cost remains a challenge. Optimizations such as input truncation and the use of lighter models like TinyBERT are discussed to improve efficiency. The findings reveal that some hotels consistently receive positive reviews, while others show improvement or maintain stable ratings.**

I. Introduction

This project aims to classify hotel reviews as positive or negative using a BERT-based text classification model, sourced from Hugging Face. By classifying reviews, we can assess customer satisfaction and help consumers choose the best hotels. The analysis will uncover sentiment trends, such as the percentage of positive/negative reviews for each hotel and shifts in sentiment over time.

Text classification, leveraging techniques like natural language processing and machine learning, extracts meaningful patterns from unstructured text. It is used in applications such as sentiment analysis, spam detection, and topic modeling. Advanced models like bi-directional LSTMs (bi-LSTMs) and capsule networks have improved text classification by capturing complex patterns, though RNNs can struggle with intricate sentence structures. Transformer-based models like BERT (Bidirectional Encoder Representations from Transformers) offer a significant advancement by capturing both forward and backward context. However, BERT is computationally expensive and resource-intensive.

II. Method

A. Problem Formulation

This project aims to classify hotel reviews to predict customer sentiment. The model takes a text review describing the customer's experience as input. All hotel reviews are stored in a CSV file. The model outputs a label of either ”positive” or ”negative.”

The hotel reviews dataset, obtained from Hugging Face, contains 516,000 reviews from 1,492 hotels. The dataset includes three string-type columns: review date, hotel name, and review text. Notably, the reviews lack punctuation such as commas, periods, or apostrophes. There are no missing values, and the average review length is 188 characters. Some of the hotels with the most reviews include the Britannia International Hotel Canary Wharf, the Strand Palace Hotel, and the Park Plaza Westminster Bridge London.

This project utilizes BERT, a model known for its bidirectional architecture and ability to capture context effectively, which is expected to achieve higher accuracy. However, there are trade-offs between computational efficiency and accuracy when using BERT. One key bottleneck is its slower training and inference times. Since the dataset does not contain sentiment labels (positive or negative), a pre-trained BERT model, DistilBERT, was used for sentiment classification. DistilBERT is a small, fast, cheap, and light Transformer model trained by distilling BERT base.

B. Methodologies

For data preprocessing, the `DistilBertTokenizer` was used to convert text into numerical tokens. This tokenizer includes 12 parameters, such as whether to lowercase the input, apply padding for batching sequences of different lengths, and split tokens during tokenization. Since this project did not tune any of these parameters, the tokenizer defaulted to lowercasing the input and performing basic tokenization before applying WordPiece, a subword segmentation algorithm.

The `DistilBertForSequenceClassification` model includes 8 parameters, such as input IDs, attention masks, and head masks. The output logits are stored as a `torch.FloatTensor` with a shape of `(batch_size, config.num_labels)`. These logits are transformed into probabilities using the softmax function for classification.

This project utilized Hugging Face Transformers for model implementation, PyTorch for training, and Pandas, Seaborn, and Matplotlib for data visualization.

III. Results

A. Data Pipeline and Model Setup

First, the data was subsetted to include only the top 8 hotels with the lowest number of reviews, given the size of the dataset and the time required for the model to classify hotel reviews. The data pipeline involved tokenizing text reviews using the DistilBERT tokenizer, performing basic tokenization before applying WordPiece, padding sequences, and feeding the data into a pre-trained DistilBERT model fine-tuned on the SST-2 (Stanford Sentiment Treebank) corpus for sentiment classification. The default fine-tuning hyperparameters were as follows: learning rate of 1e-5, batch size of 32, and number of training epochs set to 3.0.

B. Numerical Simulations

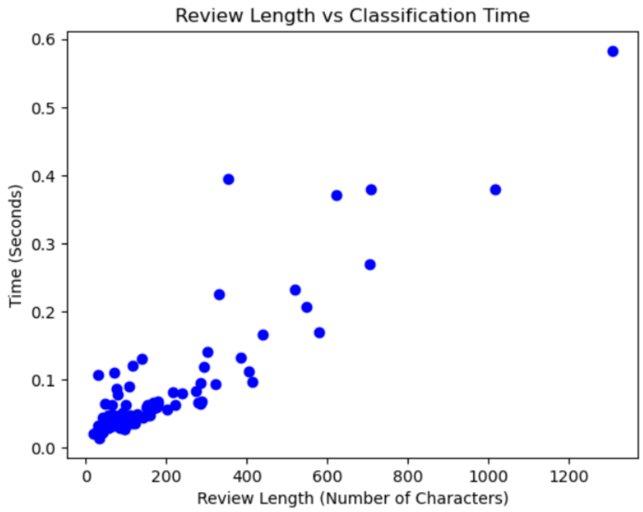

Fig. 1. Classification time (in seconds) as a function of review length (number of characters).

tokenization	inference	total time
0.000000	0.379944	0.379944
0.005138	0.134922	0.140060
0.003023	0.060438	0.063461
0.000000	0.063822	0.063822
0.000000	0.050135	0.050135
...	...	...
0.001228	0.026658	0.027886
0.001031	0.046428	0.047459
0.001026	0.032030	0.033056
0.001048	0.028736	0.029783
0.002377	0.036771	0.039148

Fig. 2. Classification time (in seconds) categorized by tokenization and inference stages.

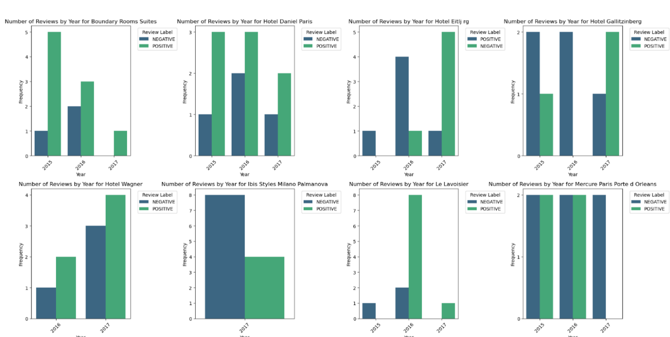

Fig. 3. Bar plot showing the distribution of positive and negative reviews by hotel and year.

C. *Interpretations*

From Fig. 1, it is evident that there is a relationship between the length of a hotel review and the time it takes for the model to classify it as positive or negative. Specifically, reviews with more characters require more time to classify, as the model must process a larger amount of information. The distribution of data points shows that most reviews in the subset are relatively short and range from 0 to 300 characters. However, there are a few reviews that are significantly longer, with one exceeding 1200 characters.

From Fig. 2, it is clear that tokenization is a quick process, as a substantial portion of the classification time is spent on inference. This can be attributed to BERT's bidirectional processing nature and the transformer architecture, which includes multiple layers of attention mechanisms.

From Fig. 3, some of the hotels that consistently have a higher proportion of positive reviews compared to negative reviews include Boundary Rooms Suites, Hotel Daniel Paris, and Hotel Wagner. For Hotel Gallitzinberg, it is noticeable that there were more negative reviews in 2015 and 2016. However, there is a significant increase in positive reviews during 2017, which may suggest that the hotel implemented changes to improve its services.

IV. Conclusion

BERT has many applications in sentiment analysis, as its bidirectional nature allows for a more nuanced understanding of sentences before classification. However, this comes at the cost of time efficiency, with most of the processing time spent on inference and predicting class labels. To optimize performance, possible solutions include truncating input sequences, using a sliding window approach to maintain sentence context, and fine-tuning the batch size. Additionally, lighter models like TinyBERT can offer faster alternatives. The data also reveals that some hotels are consistently favored by consumers, while others show signs of improvement or have maintained similar review proportions over the years.

V. GitHub Repository

https://github.com/rgu6/STAT-507/blob/main/Final%20Project/Hotel%20Reviews%20Text%20Classification.ipynb

References

[1] "515K Hotel Reviews," Hugging Face, [Online]. Available: https://huggingface.co/datasets/Ez4Real/515K_Hotel_Reviews. [Accessed: Dec. 9, 2024].

[2] "DistilBERT: A smaller version of BERT," Hugging Face, [Online]. Available: https://huggingface.co/distilbert/distilbert-base-uncased-finetuned-sst-2-english#model-details. [Accessed: Dec. 9, 2024].

[3] "DistilBertTokenizer — Transformers 4.0.0 documentation," Hugging Face, [Online]. Available: https://huggingface.co/docs/transformers/en/model_doc/distilbert#transformers.DistilBertTokenizer. [Accessed: Dec. 9, 2024].

[4] K. Taha, P. D. Yoo, C. Yeun, D. Homouz, and A. Taha, "A comprehensive survey of text classification techniques and their research applications: Observational and experimental insights," *Procedia Computer Science*, [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1574013724000480. [Accessed: Dec. 9, 2024].